

Trigonometric Fourier Series (example) for Electronics and Telecommunications

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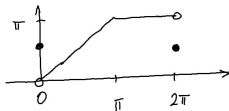
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Expand into Fourier trigonometric series

$$f(x) = \begin{cases} x, & x \in (0; \pi) \\ \pi, & x \in (\pi; 2\pi) \\ \frac{\pi}{2}, & x = 2\pi, x = 0 \end{cases}$$

Check if it satisfies Dirichlet conditions.

$$f(x) = \begin{cases} x & , x \in (0, \pi) \\ \pi & , x \in (\pi, 2\pi) \\ \frac{\pi}{2} & , x = 2\pi, x = 0 \end{cases}$$



$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

where $a_0 = \frac{1}{\pi} \left[\int_0^{\pi} x \, dx + \int_{\pi}^{2\pi} \pi \, dx \right] = \frac{1}{\pi} \left[\frac{x^2}{2} \Big|_0^{\pi} + \pi x \Big|_{\pi}^{2\pi} \right] =$

$$= \frac{1}{\pi} \left(\frac{\pi^2}{2} - 0 \right) + (2\pi - \pi) = \frac{3}{2}\pi \quad (5 \text{ pts})$$

Next we shall continue with a_n and b_n .

$$a_n = \frac{1}{\pi} \left[\int_0^{\pi} x \cos nx \, dx + \int_{\pi}^{2\pi} \cos nx \, dx \right] =$$

$$= \frac{1}{\pi} \cdot \frac{1}{n^2} [(-1)^n - 1] \quad (5 \text{ pts})$$

$$I_1 = \int x \cos nx \, dx = \left| \begin{array}{l} u = x \\ v' = \cos nx \end{array} \right| \begin{array}{l} u' = 1 \\ v = \frac{1}{n} \sin nx \end{array} =$$

$$= \frac{x}{n} \sin nx - \int \frac{1}{n} \sin nx \, dx = \frac{x}{n} \sin nx + \frac{1}{n^2} \cos nx$$

and definite integral

$$\int_0^{\pi} x \cos nx \, dx = \frac{\pi}{n} \sin n\pi - 0 + \frac{1}{n^2} \cos n\pi - \frac{1}{n^2} = \frac{1}{n^2} [(-1)^n - 1]$$

$$I_2 = \int_{\pi}^{2\pi} \cos nx \, dx = 0$$

$$b_n = \frac{1}{\pi} \left[\underbrace{\int_0^{\pi} x \sin nx \, dx}_{I_1} + \underbrace{\int_{\pi}^{2\pi} \sin nx \, dx}_{I_2} \right] =$$

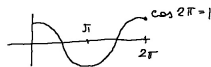
$$= \frac{1}{\pi} \left[-\frac{\pi}{n} (-1)^n - \frac{\pi}{n} [1 - (-1)^n] \right] = -\frac{1}{n} \quad (5 \text{ pts})$$

$$I_1 = \int_0^{\pi} x \sin nx \, dx = \left| \begin{array}{l} u = x \\ v' = \sin nx \end{array} \right| \begin{array}{l} u' = 1 \\ v = -\frac{1}{n} \cos nx \end{array} \Bigg|_0^{\pi} =$$

$$= -\frac{x}{n} \cos nx + \frac{1}{n} \int \cos nx \, dx = -\frac{x}{n} \cos nx + \frac{1}{n^2} \sin nx$$

$$\int_0^{\pi} x \sin nx \, dx = -\frac{\pi}{n} \underbrace{\cos n\pi}_{(-1)^n} - 0 + \frac{1}{n^2} \underbrace{\sin n\pi}_0 - 0 = -\frac{\pi}{n} (-1)^n$$

$$I_2 = \int_{\pi}^{2\pi} \sin nx \, dx = -\frac{\pi}{n} \cos nx \Bigg|_{\pi}^{2\pi} = -\frac{\pi}{n} [1 - (-1)^n]$$

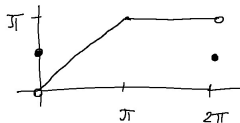


$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

where $a_0 = \frac{3\pi}{2}$, $a_n = \begin{cases} 0, & n \text{ even} \\ -\frac{2}{n^2\pi}, & n \text{ odd} \end{cases}$, $b_n = -\frac{1}{n}$

$$f(x) = \frac{3\pi}{4} - \frac{2}{\pi} \left[\cos x + \frac{1}{3^2} \cos 3x + \frac{1}{5^2} \cos 5x + \dots \right] - 1 \cdot \left[\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \frac{1}{4} \sin 4x + \dots \right] \quad (5 \text{ pts})$$

Let's check if f



satisfies Dirichlet conditions on $\langle 0, 2\pi \rangle$

- f is piecewise monotonic
- f is cont apart from finite number of discontinuities $x_0 = 2\pi$ of the first kind

$$\lim_{x \rightarrow 2\pi^-} f(x) = \pi$$

- at each pt $f(x) = \frac{f_+(x) + f_-(x)}{2}$

- $f(0) = f(2\pi) = \frac{\pi}{2}$

(5 pts)